

Assignment 1

Name: _____

Student ID: _____

6 problems. Show all of your work.

Problem 1.Show that $\tanh^2(x) + \operatorname{sech}^2(x) = 1$, for all real numbers x .**Problem 2.**Compute $\frac{d}{dx} 3 \sinh\left(\frac{3}{x}\right)$.**Problem 3.**Simplify $\cosh(\ln 2)$.**Problem 4.**Solve the equation $5 \sinh(x) - 3 \cosh(x) = 3$ for x .**Problem 5.**Following the method of the example in the notes that computes $\cosh^{-1}(x)$, show that the inverse hyperbolic tangent is given by

$$\tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad -1 < x < 1.$$

Problem 6.Use the substitution $x = \tanh(t)$ to calculate

$$\int \frac{dx}{1-x^2}, \quad -1 < x < 1,$$

and then use Problem 5 to write your answer with a logarithm. *Hint:* $\frac{d}{dt} \tanh(t) = \operatorname{sech}^2(t)$, and $1 - \tanh^2(t) = \operatorname{sech}^2(t)$.